

QCE Chemistry Units 3&4

Test 3: Oxidation and reduction

- Redox reactions
- Electrochemical cells

Student's Name: _____

Teacher's Name: _____

General instructions

- Write your name and your teacher's name in the space provided above on this page.
 - A formula and data booklet is provided.
 - Unless otherwise indicated, the diagrams in this booklet are not drawn to scale.
 - Answer all questions in the spaces provided.
- Test time – 45 minutes
 - Total marks – 40 marks

SECTION 1**Instructions**

- Choose the best answer for Questions 1–10.
 - This section has 10 questions and is worth 10 marks.
 - Circle A, B, C or D.
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QUESTION 1

Which of the following is **not** a redox reaction?

- (A) $\text{Mg(s)} + 2\text{HCl(aq)} \rightarrow \text{MgCl}_2\text{(aq)} + \text{H}_2\text{(g)}$
- (B) $\text{HCl(aq)} + \text{NaOH(aq)} \rightarrow \text{NaCl(aq)} + \text{H}_2\text{O(l)}$
- (C) $\text{C(s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$
- (D) $4\text{Fe(s)} + 3\text{O}_2\text{(g)} \rightarrow 2\text{Fe}_2\text{O}_3\text{(s)}$

QUESTION 2

Identify the substance that shows the species in **bold** having an oxidation state of +3.

- (A) **Al**
- (B) **Fe**₂O₃
- (C) **Na**Cl
- (D) **Fe**(OH)₂

QUESTION 3

Identify the half-equation for the oxidation of magnesium atoms.

- (A) $\text{Mg(s)} + 2\text{e}^- \rightarrow \text{Mg}^{2-}\text{(aq)}$
- (B) $\text{Mg(s)} + \text{e}^- \rightarrow \text{Mg}^-\text{(aq)}$
- (C) $\text{Mg(s)} \rightarrow \text{Mg}^{2+}\text{(aq)} + 2\text{e}^-$
- (D) $\text{Mg(s)} \rightarrow \text{Mg}^+\text{(aq)} + \text{e}^-$

QUESTION 4

In an electrochemical cell, electrons move from the

- (A) anode to the cathode via an external circuit.
- (B) cathode to the anode via an external circuit.
- (C) anode to the cathode via the salt bridge.
- (D) cathode to the anode via the salt bridge.

QUESTION 5

Reduction involves

- (A) an increase in the oxidation state.
- (B) a loss of electrons.
- (C) a gain of electrons.
- (D) a loss of hydrogen.

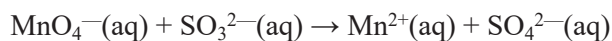
QUESTION 6

An atom in group 7 of the periodic table is likely to have an oxidation state of

- (A) -7
- (B) -1
- (C) +1
- (D) +7

QUESTION 7

Consider the equation shown.



Identify the reducing agent.

- (A) MnO_4^-
- (B) SO_3^{2-}
- (C) Mn^{2+}
- (D) SO_4^{2-}

QUESTION 8

Which of the following substances has a hydrogen atom with an oxidation state of -1?

- (A) water (H_2O)
- (B) methane (CH_4)
- (C) sodium hydride (NaH)
- (D) ethanol ($\text{C}_2\text{H}_5\text{OH}$)

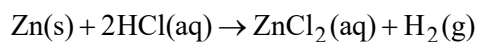
QUESTION 9

In hydrogen peroxide (H_2O_2), the oxidation state of oxygen is

- (A) -2
- (B) -1
- (C) +1
- (D) +2

QUESTION 10

The equation shows the reaction between zinc (Zn) and hydrochloric acid (HCl).



Which of the following statements is correct?

- (A) The reaction is not a redox reaction.
- (B) Zn is oxidised and chlorine (Cl) is reduced.
- (C) Zn is oxidised and hydrogen (H) is reduced.
- (D) The oxidation state of Zn changes from +2 to +1.

SECTION 2**Instructions**

- This section has nine questions and is worth 30 marks.

QUESTION 11 (1 mark)

Explain why oxidation and reduction occur simultaneously.

QUESTION 12 (3 marks)

Identify three differences between an electrolytic cell and a galvanic cell.

Electrolytic cell	Galvanic cell

QUESTION 13 (1 mark)

Determine the oxidation state of cobalt in CoCl_4^{2-} .

QUESTION 14 (5 marks)

When sodium bromide reacts with chlorine gas, a redox reaction occurs.

- a) Write the balanced oxidation half-equation for the reaction. *[1 mark]*

- b) Write the balanced reduction half-equation for the reaction. *[1 mark]*

- c) Using your answers for 16a) and b), write the complete balanced equation for the reaction. *[2 marks]*

- d) Identify the oxidising agent in the reaction. *[1 mark]*

QUESTION 15 (2 marks)

A student wishes to determine the order of the reducing strengths of three metals: A, B and C. Metal A does not react with a solution that contains ions of metal B. Metals A and B react with a solution that contains ions of metal C.

List the metals in order of increasing reducing strength.

QUESTION 16 (4 marks)

- a) Determine the oxidation number of manganese in the following species.

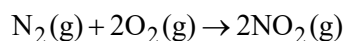
i) MnO_4^{2-} [1 mark]

ii) Mn_2O_3 [1 mark]

- b) Determine whether MnO_4^{2-} or Mn_2O_3 would be the stronger oxidising agent. Justify your answer. [2 marks]

QUESTION 17 (6 marks)

Nitrogen gas reacts with oxygen gas according to the equation shown.



- a) Using oxidation states, determine which species is undergoing oxidation and which species is undergoing reduction. Show your working [4 marks]

Species undergoing oxidation = _____ Species undergoing reduction = _____
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- b) Identify the reducing agent and oxidising agent. [2 marks]
